

Research Question	Data / Situation	Statistical Test	H_0	H_1	Test Statistic
Is a population mean different from a specified value when population σ is known?	One population, quantitative data, σ known	One-Sample Z-Test for Mean	$\mu = \mu_0$	$\mu \neq \mu_0$	Z
Is a population mean different from a specified value when population σ is unknown?	One population, quantitative data, σ unknown	One-Sample t-Test	$\mu = \mu_0$	$\mu \neq \mu_0$	t
Is a population proportion different from a specified value?	One population, binary/categorical outcome	One-Proportion Z-Test	$p = p_0$	$p \neq p_0$	Z
Is a population variance different from a specified value?	One population, quantitative data	Chi-Square Test for Variance	$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$	χ^2
Are the means of two independent populations different?	Two independent groups, quantitative outcome	Independent Two-Sample t-Test	$\mu_1 - \mu_2 = 0$	$\mu_1 - \mu_2 \neq 0$	t
Are the variances of two populations different?	Two populations, quantitative data	F-Test for Two Variances	$\sigma_1^2 / \sigma_2^2 = 1$	$\sigma_1^2 / \sigma_2^2 \neq 1$	F
Are the proportions of two populations different?	Two independent populations, binary/categorical outcome	Two-Proportion Z-Test	$p_1 - p_2 = 0$	$p_1 - p_2 \neq 0$	Z
Are the means of 3 or more independent populations different?	3+ independent groups, quantitative outcome	One-Way ANOVA	$\mu_1 = \mu_2 = \dots = \mu_k$	Not all population means are equal	F
Are the means of 3 or more related treatments different?	Same blocks measured under all treatments	Randomized Block ANOVA	$\mu_1 = \mu_2 = \dots = \mu_k$	Not all treatment means are equal	F
Does one categorical variable follow a specified distribution?	One categorical variable; observed vs. expected frequencies	Chi-Square Goodness-of-Fit Test	$p_i = p_{i0}$ for all categories	At least one $p_i \neq p_{i0}$	χ^2
Are two categorical variables associated?	Two categorical variables in a contingency table	Chi-Square Test of Independence	The two variables are independent	The two variables are associated	χ^2